Assistive Robtics

Reward Design and Task Geometry

Our wiping task is not just "touch the arm." The policy should

1. first reach the correct arm region,
2. then maintain stable contact,
3. clear the remaining target points,
4. and ideally do so through a smooth directional wiping motion rather than isolated point poking.

Because of this, the reward is built as a combination of

- distance shaping,
- action regularization,
- target-clear reward,
- near-target contact reward,
- force-band shaping,
- directional sweep reward,
- and a local directional-window reward.

Arm-region and target-point selection

Before discussing the reward itself, it is useful to clarify what the target points are.

The environment first generates target points on the human arm, separately for the

`upperarm` and `forearm`.

These points correspond to the small visible targets in the simulator (the purple points/markers in rendering).

Then, for a chosen splitting axis, these arm targets are divided into two halves:

`front` and `back`.

This creates four possible active wiping regions:

`upperarm_front`, `upperarm_back`, `forearm_front`, `forearm_back`.

In code, this is represented by dictionaries of the form

`target_pos_world_dict[arm_side]`,

where `arm_side` specifies which region is active in the current episode.

Thus, the task is not to wipe the whole arm at once, but to wipe a selected subset of target points on one region of the arm. This is important because the reward and the directional wiping logic should align with the currently active target cloud.

Intuition. The purple points are the local dirt/cleaning targets. The policy is successful when it clears these points from the currently selected arm region. So the reward should not only encourage contact with the arm, but specifically contact with these remaining target points.

Overall reward formula

At time step t, the environment returns a scalar reward

$$r_t = w_d\, r_t^{\mathrm{dist}} + w_a\, r_t^{\mathrm{act}} + w_c\, r_t^{\mathrm{clear}} + w_n\, r_t^{\mathrm{near}} + w_f\, r_t^{\mathrm{force}} + r_t^{\mathrm{sweep}} + r_t^{\mathrm{window}}, \quad (1)$$

where

$$w_d = \texttt{distance\_weight}, \quad w_a = \texttt{action\_weight}, \quad w_c = \texttt{wiping\_reward\_weight}, \quad w_n = \texttt{near\_target\_weight}$$

So in words, the total reward is:

> distance-to-task-region reward + action penalty + actual target clearing reward + near-target contact shaping + contact-force shaping + directional sweep shaping + local active-window shaping.

1. Distance term

The environment has two practical phases:

1. a **reach phase**, where the tool should move toward the arm,
2. a **wipe phase**, where the tool should remain close to the uncleared targets.

Let d_t^{arm} denote the minimum distance between the tool and the arm. Then in reach mode,

$$r_t^{\mathrm{reach}} = -d_t^{\mathrm{arm}}. \quad (2)$$

Once wiping is active, the reward instead uses the mean distance from the tool to the remaining feasible targets. Let

$$p_t^{\mathrm{tool}}$$

denote the tool position, and let

$$\{p_{i,t}^{\text{target}}\}_{i=1}^{N_t}$$

be the remaining target positions in world coordinates. Then

$$r_t^{\text{wipe-dist}} = -\frac{1}{N_t}\sum_{i=1}^{N_t} \left\| p_t^{\text{tool}} - p_{i,t}^{\text{target}} \right\|_2 . \tag{3}$$

Therefore the actual distance reward is

$$r_t^{\text{dist}} = \begin{cases} r_t^{\text{wipe-dist}}, & \text{if wipe mode is active,} \\ r_t^{\text{reach}}, & \text{otherwise.} \end{cases} \tag{4}$$

Intuition. Before contact, the robot should simply approach the arm. After contact becomes stable, it should stay close to the remaining purple target points rather than drifting away.

2. Action penalty

Let $a_t \in \mathbb{R}^m$ be the action vector. The action regularization term is

$$r_t^{\text{act}} = -\|a_t\|_2^2. \tag{5}$$

Intuition. This discourages overly large or jerky actions. It helps the policy avoid unstable motion and encourages smoother wiping behavior.

3. Target-clear reward

Let c_t denote the number of targets cleared at time step t. Then

$$r_t^{\text{clear}} = c_t. \tag{6}$$

Each cleared target contributes positively through the coefficient w_c.

Intuition. This is the main task-progress term. If the tool wipes away target points successfully, the agent gets direct reward.

4. Near-target contact reward

The environment also provides a dense binary shaping signal that checks whether the current valid contact is close to any remaining target. Define

$$n_t^{\text{near}} = \begin{cases} 1, & \text{if a valid contact is within } \texttt{near\_target\_dist} \text{ of any remaining target,} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Then

$$r_t^{\text{near}} = n_t^{\text{near}}. \tag{8}$$

Intuition. A step may not clear a target yet, but if the tool is contacting near the remaining purple points, that is still useful progress. This term gives denser feedback than sparse clearing alone.

5. Force-band shaping

Let F_t be the total normal contact force for valid tool–arm contacts. The environment uses a preferred force band

$$F_{\min} = \texttt{contact\_band\_min}, \qquad F_{\max} = \texttt{contact\_band\_max}.$$

Then the force reward is

$$r_t^{\text{force}} = \begin{cases} 0, & F_t \le \varepsilon \text{ and wipe mode is active,} \\ -1, & F_t \le \varepsilon \text{ and wipe mode is inactive,} \\ -(F_{\min} - F_t), & 0 < F_t < F_{\min}, \\ 0, & F_{\min} \le F_t \le F_{\max}, \\ -(F_t - F_{\max}), & F_t > F_{\max}, \end{cases} \tag{9}$$

where ε is a small threshold used numerically to detect almost-zero force.

Intuition. If contact is too weak, wiping is ineffective. If contact is too strong, it may be unsafe or unrealistic. So this term encourages useful and moderate sustained contact.

6. Why we added directional wiping logic

The earlier reward already encouraged reaching, contact, and clearing. However, this still allowed a behavior that looked like:

> go near a point, clear locally, and keep harvesting nearby targets without a clearly visible wipe pass.

In other words, the agent could get decent reward by local contact around individual points, even if the motion was not visibly sweeping across the arm.

To address this, we introduced a 1D coordinate along the active arm-region target cloud and used it to encourage directional wiping.

7. Arm-axis coordinate and sweep reward

For the currently active arm region (for example `upperarm_front` or `forearm_back`), the world-space target points are used to estimate an arm-axis direction. Let

$$s_t \in [0, 1]$$

denote the normalized projection of the current tool/contact position onto this axis.

Thus:

- $s_t \approx 0$ means the tool is near one end of the active target strip,
- $s_t \approx 1$ means the tool is near the other end.

We also maintain a persistent desired pass direction

$$d_t^{\text{pass}} \in \{+1, -1\},$$

where $+1$ means moving forward along the current axis and -1 means reverse/return pass.

Define the raw change in arm-coordinate as

$$\Delta s_t = s_t - s_{t-1}, \tag{10}$$

and the direction-aligned progress as

$$\Delta s_t^{\text{dir}} = d_t^{\text{pass}}\, \Delta s_t. \tag{11}$$

Then the sweep reward is

$$r_t^{\text{sweep}} = \begin{cases} k_f \min(\Delta s_t^{\text{dir}}, \Delta_{\max}), & \Delta s_t^{\text{dir}} > \epsilon_s, \\ -\gamma_t\, k_b \min(-\Delta s_t^{\text{dir}}, \Delta_{\max}), & \Delta s_t^{\text{dir}} < -\epsilon_s, \\ 0, & |\Delta s_t^{\text{dir}}| \le \epsilon_s, \end{cases} \tag{12}$$

where:

- ϵ_s is a deadband for small jitter,
- $\Delta_{\max}$ caps the effect of one step,
- k_f and k_b are forward and backward coefficients,
- $\gamma_t \in (0, 1]$ reduces backward penalty briefly after a recent clear.

Intuition. This term says:

> if you are in a wipe pass, move along the arm in the intended direction; do not just hover and poke one spot.

Small local jitter is tolerated, but meaningful forward progress is rewarded.

8. Directional local-window reward

Even with a sweep reward, the policy may still clear points somewhat loosely if the environment allows nearby targets to disappear around the contact.

To make the behavior more spatially coherent, we define an *active directional window* along the arm-axis.

Let the current arm-axis coordinate be s_t. Then for a window width Δ_w:

- if $d_t^{\text{pass}} = +1$, the active window is approximately

$$[s_t,\ s_t + \Delta_w],$$

- if $d_t^{\text{pass}} = -1$, the active window is approximately

$$[s_t - \Delta_w,\ s_t].$$

This means that at each step, the policy is encouraged to care most about the local strip of targets currently ahead in the wipe direction.

Let

$$I_t^{\text{win-near}} \in \{0, 1\}$$

indicate whether the current valid contact is near at least one target in this active window, and let

$$C_t^{\text{win}} \in \mathbb{N}$$

denote the number of clears that occurred inside the active window. Then

$$r_t^{\text{window}} = \mathbf{1}\{\text{wipe mode active and contact latched}\} \left(\alpha\, I_t^{\text{win-near}} + \beta\, C_t^{\text{win}}\right), \tag{13}$$

for coefficients $\alpha, \beta \geq 0$.

Intuition. This term encourages local, sequential wiping. Instead of just touching any target anywhere nearby, the policy is nudged toward the targets in the currently active strip of the wipe pass.

9. Full reward again, with interpretation

Putting everything together,

$$r_t = w_d\, r_t^{\text{dist}} + w_a\, r_t^{\text{act}} + w_c\, r_t^{\text{clear}} + w_n\, r_t^{\text{near}} + w_f\, r_t^{\text{force}} + r_t^{\text{sweep}} + r_t^{\text{window}}. \tag{14}$$

Each part has a specific role:

- r_t^{dist}: go toward the correct arm region and stay near remaining targets,
- r_t^{act}: avoid excessive or unstable motion,
- r_t^{clear}: directly reward actual cleaning progress,
- r_t^{near}: provide dense signal before full clears,
- r_t^{force}: maintain safe and useful contact,
- r_t^{sweep}: encourage visible directional wiping motion,
- r_t^{window}: encourage local sequential progress within the active wipe strip.

10. Practical interpretation

In simple words, the reward is trying to teach the policy:

> First reach the selected arm region. Then stay in stable contact, remain near the remaining purple targets, use reasonable force, clear targets, and do it through a directional wipe over the active local strip rather than random point touching.

This is exactly why we added the sweep and window terms on top of the original contact/clearing reward: the original setup could already clear many points, but the motion was not always visibly wipe-like. The new terms are meant to bridge that gap between *task success* and *human-interpretable wiping behavior*.